CST8507 Assignment 2 — Proposal

AI Textbook Q&A System:
A RAG-Based Educational Question Answering System
with Deep Source Tracing

Team Details

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Topic and Motivation

We will build an **Educational Question Answering** system that answers questions about AI/ML concepts using a curated collection of canonical textbooks as the knowledge base.

Motivation: Students studying AI, machine learning, and NLP face the challenge of navigating across many textbooks to find relevant information. Our system allows users to ask natural language questions and receive accurate, grounded answers with deep source tracing — not just citing a book title, but pinpointing the exact page and region where the answer was found. Users can click on any source reference to see the original PDF page with the relevant area highlighted. This promotes transparency and efficient learning.

Dataset

Our knowledge base consists of 30+ canonical textbooks in PDF format, covering:

Domain	Books	Examples
Machine Learning	7 books	ISLR, ESL, PRML, Deep Learning (Goodfellow), PML (Murphy)
Mathematics	5 books	MML (Deisenroth), Convex Optimization (Boyd), Information Theory (MacKay)
NLP	3 books	SLP3 (Jurafsky), Intro to IR (Manning), NLP Notes (Eisenstein)
Reinforcement Learning	1 book	RL: An Introduction (Sutton & Barto)
Computer Vision	1 book	Computer Vision (Szeliski)
Python	3 books	Fluent Python (Ramalho), Python Cookbook (Beazley), Think Python (Downey)

Preprocessing Pipeline

1. **Layout analysis** using MinerU (magic-pdf) with DocLayout-YOLO (GPU-accelerated) to detect document regions (text, tables, formulas, figures) with bounding box coordinates

2. **Specialized extraction** per content type: text → plain text, tables → HTML, formulas → LaTeX, figures → image + caption

3. **Intelligent chunking** with metadata (book title, chapter, page number, bounding box coordinates)

4. **Dual indexing:** SQLite with FTS5 full-text search index (BM25) + ChromaDB vector store with sentence-transformers embeddings (all-MiniLM-L6-v2)

5. **PageIndex tree generation** — hierarchical table-of-contents tree per book for reasoning-based retrieval

Source: All textbooks are publicly available open-access editions (e.g., from authors' websites or institutional repositories) or legally obtained educational copies already in our possession. Total dataset size: ~500MB of PDF documents.

Approach

RAG Architecture: We adopt a hybrid retrieval architecture that combines four complementary retrieval methods, fused via Reciprocal Rank Fusion (RRF), to maximize retrieval quality across different query types.

1. **Document Processing:** MinerU (magic-pdf) for layout-aware PDF parsing — uses DocLayout-YOLO to detect 10 categories of document elements, preserving bounding box coordinates

2. **Chunking:** Layout-aware chunking that keeps tables and formulas intact, with source metadata (book, page, section, bbox)

3. **Four Retrieval Methods:**

① **SQLite FTS5 Keyword Search (BM25):** Full-text search using SQLite's built-in FTS5 extension with BM25 ranking — fast exact keyword matching

② **ChromaDB Vector Search (Semantic):** Embedding-based retrieval using sentence-transformers (all-MiniLM-L6-v2) + ChromaDB — understands synonyms and semantic similarity

③ **PageIndex Tree Search (LLM Reasoning):** Build a hierarchical TOC tree per textbook, then use the LLM to navigate top-down to find relevant sections

④ **Metadata Filter Search (Structured):** Structured filtering by book title, chapter, page number, and content type

4. **Result Fusion:** Reciprocal Rank Fusion (RRF) combines results from all four methods into a single ranked list

5. **LLM:** Ollama with qwen2.5:0.5b model (~0.4GB memory footprint)

6. **UI:** Streamlit interface with clickable source references that highlight the exact region on the original PDF page

Key Feature — Deep Source Tracing: Each answer includes references to the exact book, chapter, page, and spatial location. Clicking a reference renders the original PDF page with the source region highlighted in a yellow bounding box, enabling users to instantly verify the answer against the original material.

Plan of Work

Week	Milestone
Week 1 (Mar 3–9)	MinerU setup, PDF layout analysis, SQLite + ChromaDB schema, batch processing
Week 2 (Mar 10–16)	Chunking + embedding, FTS5 + ChromaDB indexing, PageIndex tree building
Week 3 (Mar 17–23)	Four retrieval methods + RRF fusion, Ollama integration, RAG pipeline
Week 4 (Mar 24–30)	Streamlit UI with source tracing, evaluation (20 test questions)
Week 5 (Mar 31–Apr 3)	ROS 2 integration (Part 2), final report, presentation

Evaluation Method: We will prepare 20 domain-specific questions spanning multiple textbooks, run the system on each, and manually assess correctness using a 3-level scale (1 = correct, 0.5 = partially correct, 0 = incorrect). For each question, we will also list the top 3 retrieved chunks and mark their relevance. The final accuracy score will be the average across all 20 questions.

Expected Results

- A functional RAG-based Q&A system that answers AI/ML educational questions with >80% accuracy
- Deep source tracing with clickable references that highlight the exact region on original PDF pages
- Layout-aware processing that correctly handles tables, formulas, and figures without losing structure
- A responsive Streamlit UI for interactive querying

Why Is It Interesting?

1. **Deep source tracing —** Goes beyond basic RAG by providing pixel-level traceability to original documents, not just page references

2. **Layout-aware parsing —** Uses YOLO-based document layout detection to preserve tables, formulas, and figures that traditional text extraction would break

3. **Practical utility —** Directly useful for students studying AI/ML across multiple textbooks

4. **Scalable —** The same architecture can be applied to any domain of knowledge

5. **Low resource —** Runs entirely locally with <1.5GB model, suitable for deployment on constrained hardware (robots)